Spore-informed process-based modelling of grapevine downy mildew onset in subtropical production

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Abstract

BACKGROUND: Preventive control of grapevine downy mildew requires infection risk to be assessed from inoculum availability, host susceptibility and weather. In humid subtropical vineyards, double cropping maintains green host

tissue across production cycles and can sustain asexual sporangia beyond the seasonal reset assumed by oospore-based models. FSIM estimates the first visible symptom-onset window during the first crop growing season, and FSIM-S assesses the additional value of field spore observations.

RESULTS: Across six Guangxi site–years, airborne *Plasmopara viticola* sporangia were detected 9–72 days before the first visible symptoms. FSIM links cohort-based sporangium production with dispersal, deposition, zoospore release, infection establishment and incubation to estimate the symptom-onset window. Validation MAE was 5.67 days for FSIM and 3.83 days for FSIM-S. FSIM-S incorporates an observation-derived relative spore-pressure field before dispersal and improved retrospective onset estimation. Regional FSIM-S reconstruction produced mean onset dates of DOY 195–198, with most estimates between DOY 180 and 210, and indicated an average of 3.0 potentially avoidable sprays per season (median 2.7) relative to the fixed-calendar programme.

CONCLUSION: Representing recurrent sporangial inoculum before symptom appearance addresses a modelling gap in subtropical vineyards. FSIM-S improves onset estimation and supports process-based timing of preventive fungicide applications during the first crop growing season. Regional onset maps and spray-saving estimates support spatially targeted, risk-based disease management across Guangxi's double-cropping vineyards.

Keywords: Agroclimatic modelling; Airborne sporangia; Disease-onset estimation; Fungicide timing; Subtropical viticulture

Introduction

Grapevine downy mildew, caused by *Plasmopara viticola*, threatens production in humid subtropical regions, where warm, rainy conditions coincide with the emergence of susceptible leaves and development of inflorescences (Koledenkova et al., 2022; Peng et al., 2024). Epidemics can impair canopy photosynthesis, damage inflorescences and berries, and cause substantial yield loss (Nogueira Júnior et al., 2019). Disease control depends heavily on repeated preventive fungicide applications (Gessler et al., 2011; Caffi et al., 2010). Timing based on calendar schedules, grower experience or qualitative weather cues may be poorly aligned with infection risk. Understanding when inoculum can establish infection is therefore central to protecting susceptible tissue before symptoms appear. Preventive management based on infection risk requires joint consideration of inoculum availability, host susceptibility and environmental suitability (Scholthof, 2007).

Empirical models such as the "3–10 rule" and EPI use weather and phenological indicators to guide control timing (Baldacci, 1947; Strizyk, 1983; Caffi et al., 2007), but they do not resolve the biological sequence from inoculum availability to symptom appearance. Process-based models provide a more explicit representation by linking the inoculum source, spore maturation or production, dispersal, deposition, zoospore release, infection and incubation (Rossi

et al., 2008; Brischetto et al., 2020, 2021). Rossi et al. (2008) used cohorts to represent primary infection from overwintered oospores, whereas Brischetto et al. (2021) modelled sporangia produced on lesions and the subsequent infections. Together, these studies describe the complete sequence of infection processes with greater biological interpretability than empirical rules, but they begin from distinct inoculum states: germinating oospores or existing visible lesions.

These initial conditions do not fully represent humid subtropical vineyards. The long growing season and mild winters support commercial double cropping in southern China, including Guangxi. The first crop growing season is typically the summer cropping cycle; pruning and bud forcing after the first harvest then initiate a second crop growing season in winter. Two harvests per year can increase annual yield and extend the marketing season (Chen et al., 2017; Lu et al., 2022). However, successive crops also extend the availability of green grapevine tissues, allowing the pathogen to persist on living hosts and replenish airborne sporangia through repeated asexual reproduction between cropping cycles (Camargo et al., 2019). This continuity contrasts with the temperate disease cycle, in which resting oospores survive winter in the absence of susceptible host tissue and germinate to provide the initial inoculum (Gessler et al., 2011; Vercesi et al., 2010). Predominantly clonal populations, asexual persistence or limited oospore occurrence have been reported in subtropical or warm viticultural regions, including São Paulo, South Africa and most surveyed Western Australian vineyards (Camargo et al., 2019; Koopman et al., 2007; Killigrew et al., 2005). Asexual sporangia may therefore be an important source of initial inoculum in Guangxi. Models initialized from oospores or visible local lesions do not directly represent this source before symptoms appear, leaving a gap in predicting disease onset during the first crop growing season.

We developed the *First Seasonal Infection Model* (FSIM) to estimate the first visible symptom-onset window during the first crop growing season, when recurrent airborne inoculum is present. Starting from a pool of asexual inoculum, FSIM represents sporangium production, airborne dispersal, deposition, zoospore release, infection establishment and incubation. It combines the cohort structure of Rossi et al. (2008) with the sporangial processes described by Brischetto et al. (2021) to estimate the window of first symptom appearance. FSIM-S denotes the spore-constrained extension of FSIM, in which field observations of background airborne sporangia provide information on site–year variation in inoculum availability within the same mechanistic framework.

The framework was evaluated using assessments of disease onset and airborne sporangia records from six Guangxi site–years collected during 2022–2024. We assessed whether FSIM and FSIM-S reproduced the timing of the first visible symptoms during the first crop growing season and whether the spore-constrained extension improved retrospective symptom-onset estimation. By linking regional inoculum availability with symptom-onset timing, the framework provides a basis for scheduling preventive fungicide applications according to infection risk in Guangxi's subtropical vineyards.

Materials and methods

Study design and overall workflow

This field-based modelling study combined observations from two commercial vineyards and one experimental orchard in Nanning, Guilin and Hechi, Guangxi, during 2022–2024. The analysis comprised field monitoring, development of FSIM and FSIM-S, and retrospective comparison of disease-onset estimates across six site–years (Fig 1). Observations included airborne sporangia, first confirmed visible downy mildew symptoms, grapevine phenology and agronomic management records. Site-specific hourly meteorological records provided the environmental forcing for model simulations (Hersbach et al., 2020).

FSIM simulated the infection-establishment, incubation and symptom-development processes during the first crop growing season from weather, phenology and a recurrent sporangial inoculum pool, and estimated the first visible symptom-onset window. FSIM-S retained the same process chain and incorporated an observation-derived relative spore-pressure field. Both models were calibrated and evaluated with interval-censored symptom-onset observations under site–year leave-one-out cross-validation; FSIM-S was then applied to regional onset reconstruction and spray-management scenario analysis.

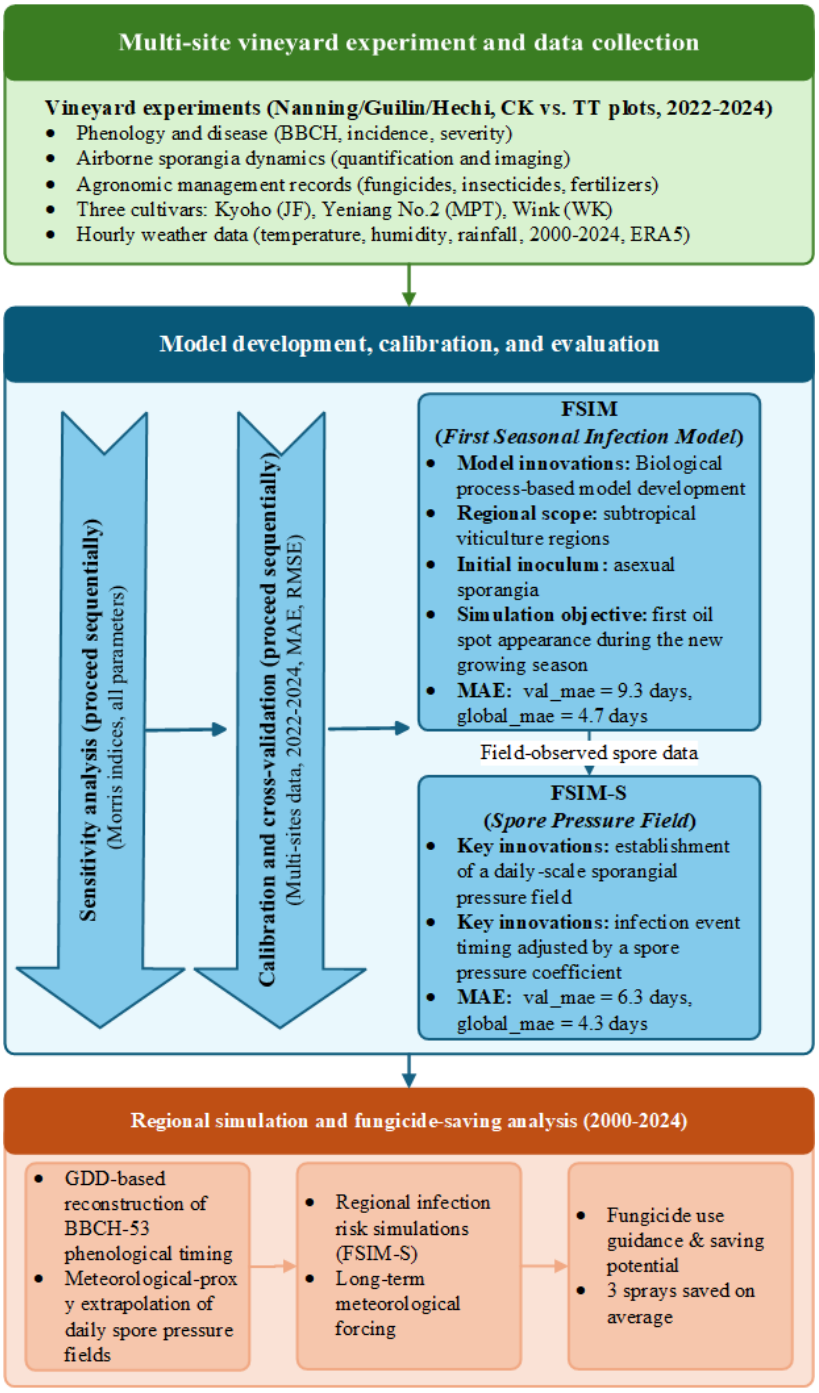


Figure 1: Overall workflow of this study. CK indicates control plots without fungicide applications, whereas TT denotes plots managed under conventional fungicide schedules.

Field experiments and data collection

Site description, climate, vineyard layout

Field experiments were conducted from 2022 to 2024 in three grape production orchards in Guangxi, China: Xing'an in Guilin, Du'an in Hechi, and Jiangnan in Nanning. The orchards were selected to represent common local production scales and cultivars, and their site characteristics are summarized in Table 1 (Fig 2a,b).

The study sites have suitable autumn–winter conditions, with relatively mild temperatures, high humidity, and recurrent rainfall (Fig 3). Orchards surrounding the experimental sites commonly use a double-cropping production system.

Under this system, vines are pruned and forced again after the first crop is harvested. The autumn crop then continues to develop through autumn and is harvested in January of the following year, allowing grape production to continue through the cool season.

At each vineyard, two management treatments were established to distinguish natural infection dynamics from management-modified disease pressure: a traditional treatment (TT), in which fungicides were applied following local management calendars, and a no-spray control (CK), in which no fungicides were applied. A completely randomized design was used, with three replicates per treatment, resulting in six observation plots per vineyard. Each plot consisted of six uniformly growing vines, which were used for subsequent phenological and disease monitoring.

Table 1 Characteristics of experimental vineyard sites in Guangxi, China.

City/County	Lat, Lon	Cultivar	Orchard type	Planting year	Row × Spacing (m)
Xing'an, Guilin	25.45°N, 110.80°E	Wink (WK)	Commercial	2018	2.5 × 2.5
Du'an, Hechi	23.83°N, 108.11°E	Yeniang No. 2 (MPT)	Commercial	2015	2.5 × 2.5
Jiangnan, Nanning	22.61°N, 108.23°E	Kyoho (JF)	Experimental	2020	3.2 × 3.0

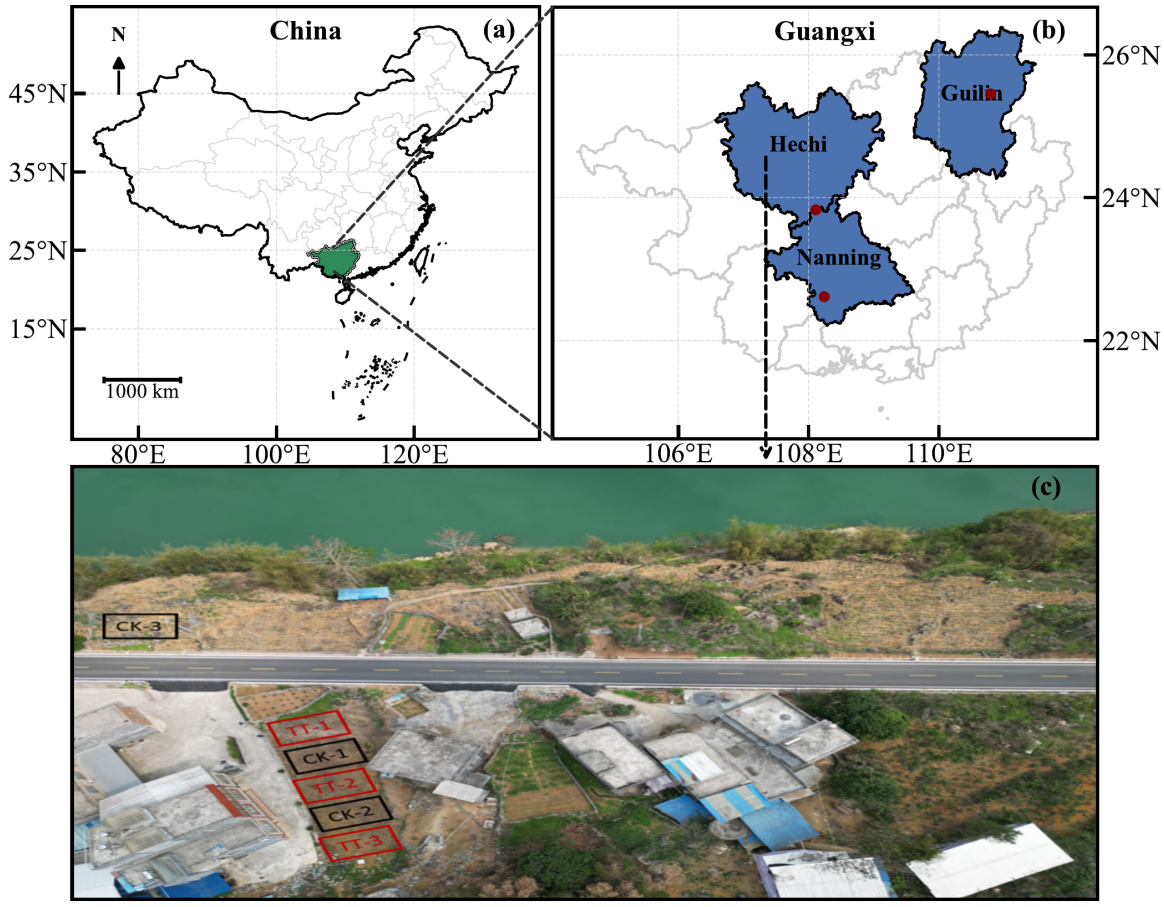


Figure 2: Study sites and field layout. (a) Location of Guangxi within China. (b) Spatial distribution of the three field experiment sites (Guilin, Hechi, and Nanning) in Guangxi Province; red circles indicate the locations of the experimental orchards. (c) Plot layout at the Du'an site in Hechi City, where black rectangles denote the three untreated control plots (CK).

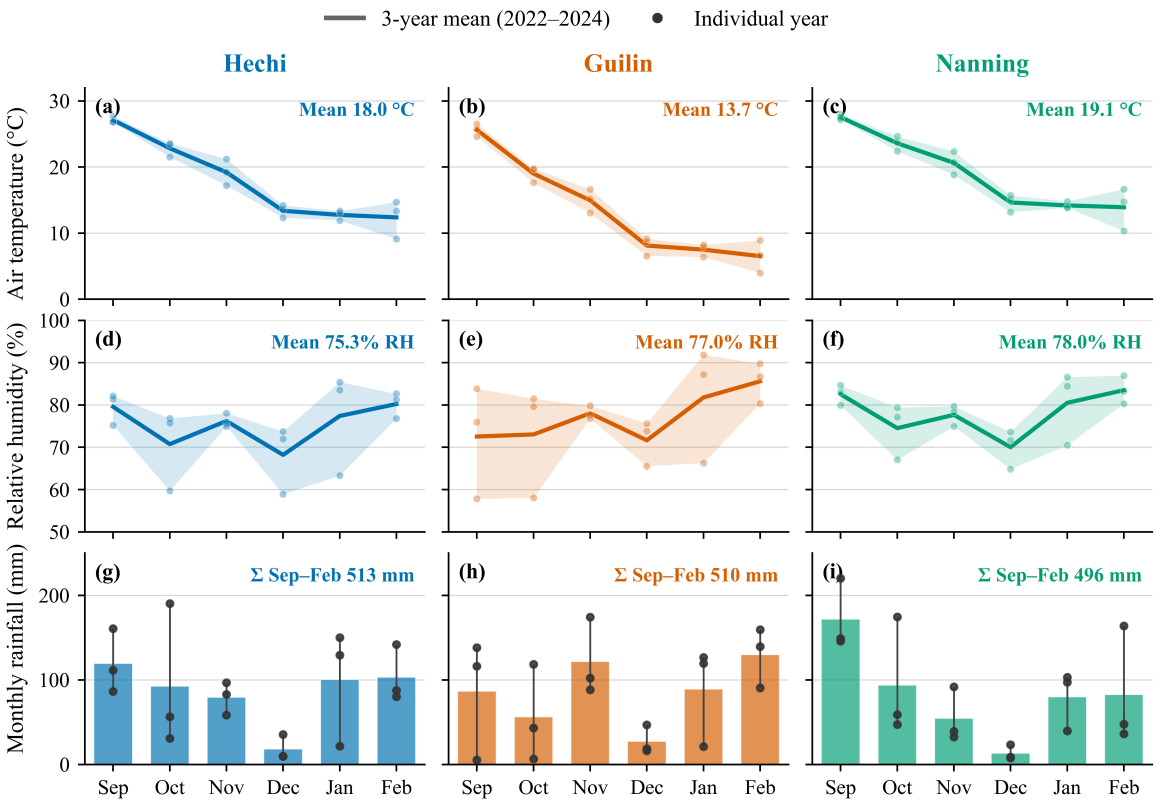


Figure 3: Monthly autumn–winter air temperature, relative humidity, and rainfall at the three study sites in Guangxi, China, summarized from 2022–2024 observations.

Disease and phenology data collection

Grapevine phenological development and downy mildew progression were monitored throughout the growing season following a standardized field observation protocol. All assessments were conducted in situ by trained technicians and recorded using a dedicated WeChat-based mobile data collection system ("Nong Qing Zhu Shou"), which provided structured interfaces for extended BBCH coding (Lorenz et al., 1995), lesion annotation, automatic time-stamping, GPS geolocation, and image capture. Records were uploaded directly to a cloud database, allowing continuous checks of data completeness and internal consistency.

Observation frequency was adjusted according to developmental stage: surveys were conducted weekly from BBCH 00 to BBCH 13 (dormancy to three unfolded leaves), and every two days thereafter as vines entered the susceptible period for downy mildew. Suspected lesions were photographed and centrally reviewed; the first confirmed oil spot in any observation plot defined the first symptom-onset date (Fig 4b).

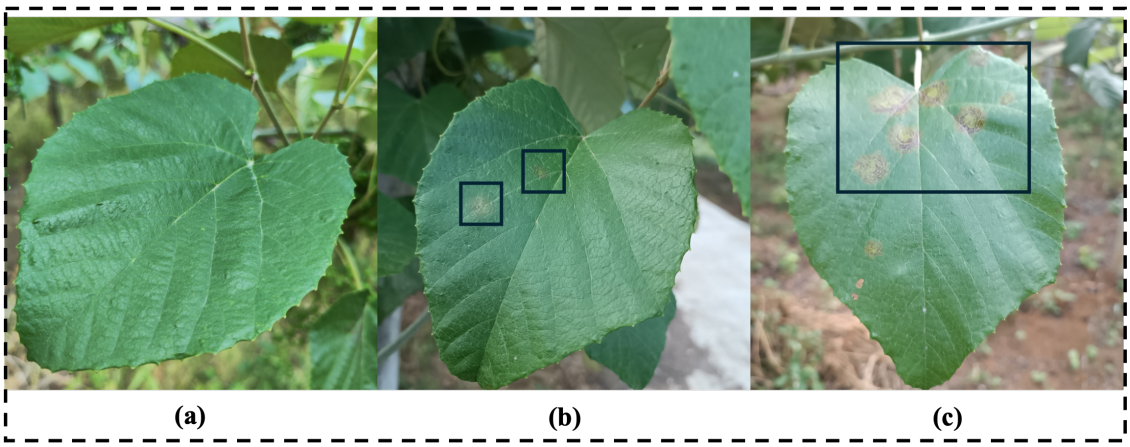


Figure 4: Downy mildew lesion emergence and development on Yeniang No. 2 (MPT) leaves in the Du'an experimental vineyard, Hechi. Panels show severity levels from a single-leaf perspective: (a) healthy; (b) 1% (post-incubation visible lesion); (c) 5%. Photos by team member Qi Tian (2023).

Airborne sporangia monitoring

At each vineyard, one passive spore trap (model TPBZ1, Zhejiang Top Cloud-Agri Technology Co., Ltd., China) was installed adjacent to CK-1, the fungicide-free control plot, to monitor the airborne dispersal of *P. viticola* sporangia. The air inlet was positioned approximately 1.5 m above ground level, matching the average grapevine canopy height. The device sampled sporangia through natural ambient airflow without powered air suction.

Sporangia were collected on glass slides coated with a thin layer of Vaseline, which were replaced every two days. Samples were sealed and transported to the laboratory of the Guangxi Academy of Agricultural Sciences for quantitative analysis. For microscopic observation, each slide was rinsed with 300 μL of sterile water, and five 30 μL subsamples were collected from different depths of the rinse solution to minimize bias associated with particle sedimentation (Fig 5b–d). Subsamples were transferred to concave glass slides for microscopic observation and imaging.

Images and resulting counts were independently verified by two trained laboratory technicians before being used in model calibration and validation (Fig 5c).

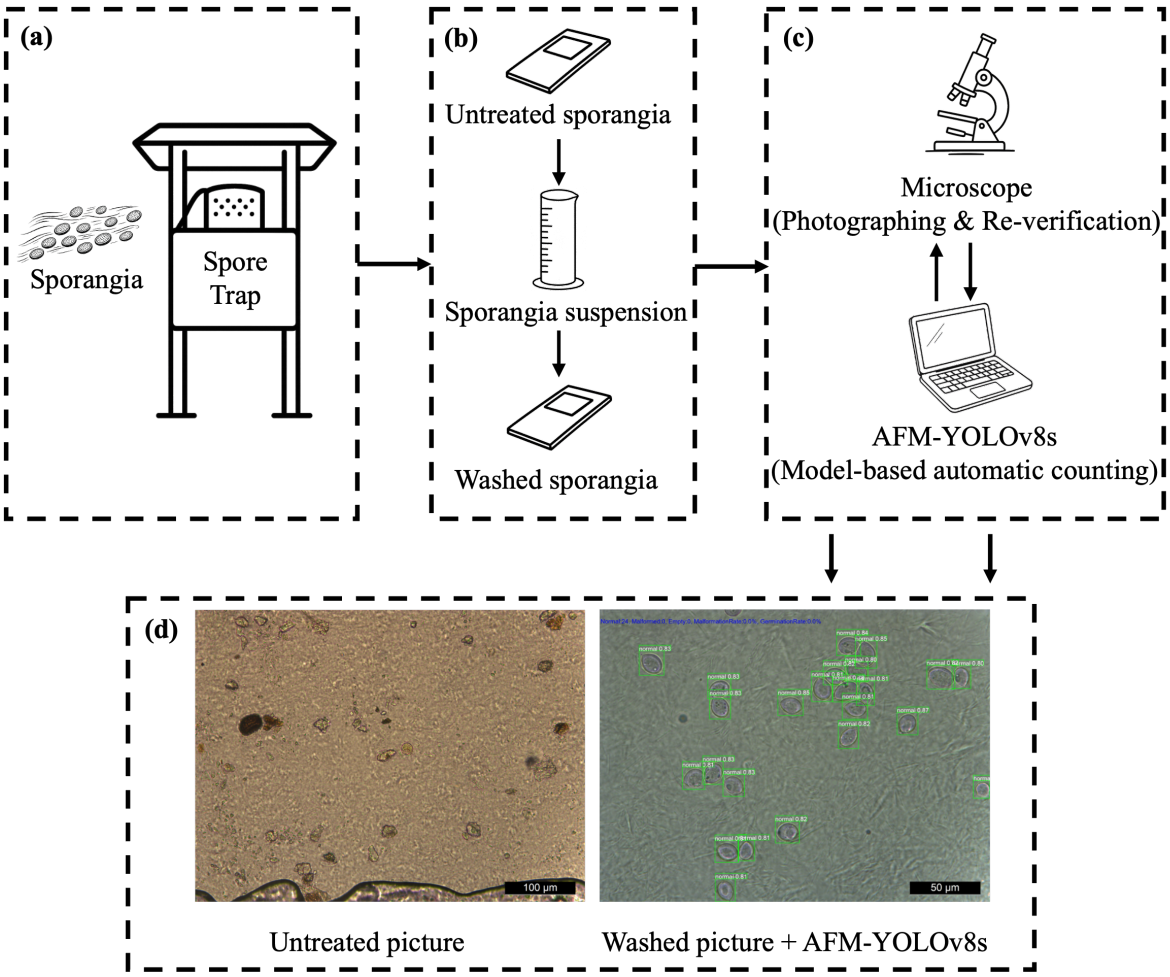


Figure 5: Schematic illustration of sporangia collection and processing in the vineyard experiment. (a) Schematic of dynamic monitoring of sporangia; (b) schematic of laboratory preprocessing; (c) schematic of sporangia image and quantity acquisition; (d) representative sporangia images obtained from the untreated sporangia slide and the washed sporangia slide.

Meteorological data

Meteorological data were retrieved from the Open-Meteo platform (https://open-meteo.com/), which provides ERA5 reanalysis products with hourly temporal resolution (Hersbach et al., 2020). For each experimental orchard, geographic coordinates were used to extract site-specific time series of air temperature (T), relative humidity (RH), wind speed (W), dew-point temperature (T_d), and precipitation (R). Vapor pressure deficit (VPD) was calculated from T and RH following Rossi et al. (2008). Because direct leaf-wetness observations were unavailable, dew-point depression (DPD) was used as a leaf-wetness proxy (Huber and Gillespie, 1992). DPD was calculated as $\mathrm{DPD} = T - T_d$; rainfall or DPD < 2.0 °C indicated wetness, DPD > 3.8 °C indicated drying, and intermediate values retained the previous wet/dry state.

Model structure and refinements

First Seasonal Infection Model (FSIM): baseline mechanistic framework

The First Seasonal Infection Model (FSIM) was developed to simulate the infection-establishment, incubation, and symptom-development chain initiated by asexual sporangia during the first crop growing season in subtropical vineyards.

Because vines in the study region commonly undergo leaf fall during January–February, FSIM simulations began on 1 February. The FSIM was constructed through structural integration and process reorganization of the Rossi primary infection model and secondary infection risk frameworks (Brischetto et al., 2021; Rossi et al., 2008).

The FSIM process chain, summarized in Fig 6, ultimately outputs the window or time interval in which the first visible lesion appears. Its components retained from the Rossi model include zoospore release and survival, infection establishment, incubation, and symptom expression, which constitute the mechanistic backbone of the infection-to-symptom chain. The secondary infection risk module was positioned upstream to describe the production of sporangia, followed by the infection chain derived from the Rossi model, including zoospore release and survival, infection establishment, incubation, and symptom expression. In this restructuring, the probability-accumulation formulation of the original secondary infection model was reformulated into a cohort-based representation describing spore production, accumulation, depletion, and release. Because the inoculum entering the first-season route was treated as an external input, transfer to the airborne state was wind-driven. Rainfall provided the deposition opportunity and the wet conditions required by subsequent survival and release. A heavy-rain wash-off adjustment was included because intense rainfall can remove spores from the air column and reduce long-range dispersal (Gobbin et al., 2005).

The cohort-based dispersal–deposition mechanism was defined as follows. The attached sporangia pool A_t was updated before transfer. When A_t exceeded the minimum threshold $A_{\min}$ and wind speed W_t exceeded $W_{\min}$, the pool was transferred to the airborne state:

$$B_{t_0} = A_{t_0},\ A_{t_0} \to 0 \tag{1}$$

where t_0 denotes the wind-transfer time and B_{t_0} the initial airborne sporangia amount. Detached-state loss started at t_0 and was updated hourly while the cohort remained airborne:

$$B_{t+1} = B_{t_0} \max\left(1 - \sum_{u=t_0}^{t} \lambda_u, 0\right), \qquad t \geq t_0, \tag{2}$$

where λ_t is the weather-dependent detached-state loss rate. An effective deposition event occurred only when the

surviving airborne amount was positive and rainfall was sufficient for deposition but below the wash-off threshold:

$$D_{t_d} = B_{t_d^-}\,\mathbf{1}\big(R_{\text{dep}} \le R_{t_d} < R_{\text{wash}}\big)\,\mathbf{1}\big(B_{t_d^-} > B_{\min}\big), \tag{3}$$

where t_d is the deposition time, R_{dep} is the effective-deposition threshold, and R_{wash} is the heavy-rain wash-off threshold. Rain below R_{dep} leaves the cohort airborne, whereas rain at or above R_{wash} removes it before effective deposition. Thus, deposition can contribute to infection only while the detached cohort remains within its survival period.

Field monitoring indicated that airborne pathogen structures captured prior to the appearance of visible lesions were predominantly sporangia (Fig 5d), providing observational support for the sporulation, dispersal, and deposition modules included in the FSIM.

Spore-constrained refinement of the *First Seasonal Infection Model* (FSIM-S)

FSIM-S was defined as an observation-constrained extension of FSIM. It retained the FSIM sequence of weather-triggered sporangium production, wind-gated transfer, rainfall-mediated deposition, detached survival, infection establishment, incubation, and symptom expression. The extension used airborne sporangium observations to constrain inoculum availability and used intervals with numeric zero counts to suppress weather-generated cohorts when no sporangia were captured. Missing entries were treated as missing and did not activate this suppression rule.

Airborne sporangia were recorded as accumulated counts over collection intervals. Let t_j denote the jth non-missing collection date and C_j its count. For an interval of duration $\Delta_j = (t_j - t_{j-1})/\text{d}$, the corresponding daily rate was calculated as

$$r(d) = \frac{C_j}{\Delta_j}, \qquad t_{j-1} < d \le t_j. \tag{4}$$

The daily rates were standardized using the mean $\bar{r}_M$ and standard deviation s_M of the measured exposure days M, smoothed with a three-day centered mean, and centered on the reference level $L = 0.5$. For day d, $W_d = \{d-1, d, d+1\} \cap \mathcal{D}$, where $\mathcal{D}$ denotes the weather calendar, the resulting centered spore-pressure signal was

$$P(d) = \frac{1}{|W_d|} \sum_{u \in W_d} \frac{r(u) - \bar{r}_M}{s_M} - L, \qquad L = 0.5. \tag{5}$$

The daily value $P(d_t)$ was assigned to each hourly timestamp t on the corresponding local calendar day. At the daily sporangium-availability time (08:00), the positive part of this signal was converted to an external background dose. The total dose entering the attached sporangium pool was defined as

$$D_t^{\text{obs}} = \beta_{\text{sp}} \max\{P(d_t), 0\}, \qquad D_t^{\text{total}} = G(d_t) D_t^{\text{weather}} + D_t^{\text{obs}}. \tag{6}$$

Here, (G(d)=0) for days covered by an interval with a numeric zero count and (G(d)=1) otherwise; missing observations therefore do not suppress weather-generated cohorts.

The combined dose then entered the FSIM attached-sporangium pool and passed through the same wind transfer, rain-mediated deposition, survival, infection-establishment, incubation, and symptom-development processes. Thus, FSIM-S constrains inoculum supply before dispersal while retaining the FSIM downstream process relations. The conceptual structure of the baseline and spore-constrained models is summarized in Fig 6.

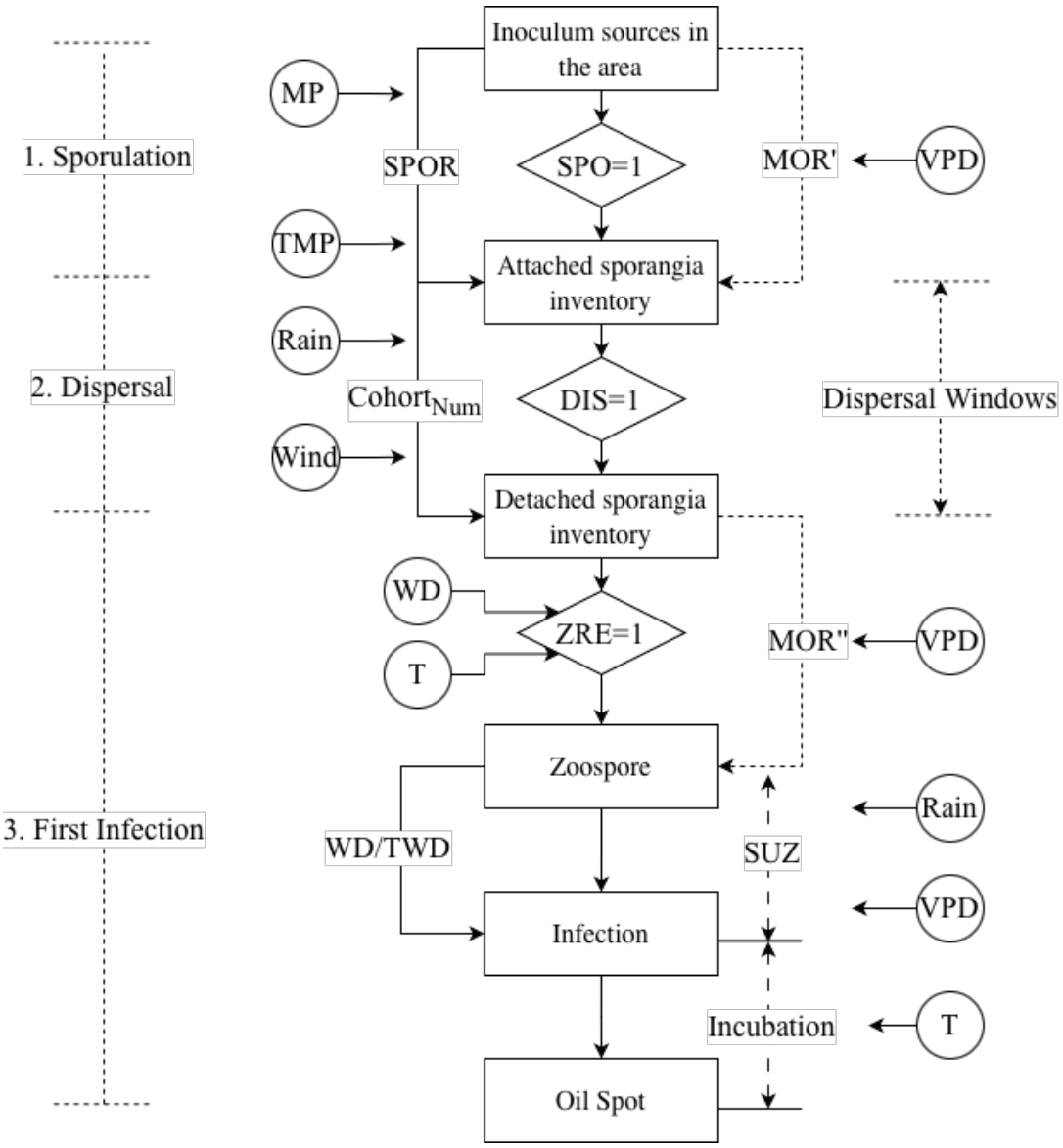


Figure 6: Conceptual diagram of the FSIM process chain. FSIM-S adds an observation-derived background sporangia dose before dispersal and suppresses weather-generated cohorts during numeric-zero observation intervals, while retaining the subsequent survival, infection-establishment, incubation and symptom-development processes.

Parameterization and sensitivity analysis

Global parameter influence was quantified for the sporangia-process route using variance-based Sobol sensitivity analysis (Sobol, 2001; Hui et al., 2022; Iooss and Lemaître, 2015). Model parameters were sampled according to the implementation settings. The observation-derived spore-pressure field and pressure-specific coefficients were held fixed, and the sensitivity analysis was used for parameter screening rather than calibration.

For each base sample size N (256, 512, and 1024), a Sobol sequence with first-, total- and second-order indices was

generated using SALib. The same site–year inputs, observed phenology and interval-based onset metric were used for every design. First-order S_1, total-order S_T, and second-order S_2 indices were estimated with 100 bootstrap resamples and 95% confidence intervals. Missing symptom-onset windows were retained in the coverage outputs and assigned the prespecified 60-day penalty only for penalized MAE; outputs with nonfinite values or zero variance were not decomposed. Stability was assessed by comparing indices and parameter rankings across the three values of N, rather than by treating larger designs as additional biological replicates. No parameter configuration was selected from the sensitivity analysis.

Table 2 Model parameters under the FSIM and FSIM-S frameworks. FSIM denotes the baseline model, whereas FSIM-S represents the extended version incorporating regulation by an observed spore-pressure field.

Parameter	Abbreviation	FSIM	FSIM-S	Unit	Description
Mp_tmp__rh_wet_night	MTWN	87.5894	83.6223	%	Night-time relative humidity threshold used to define wet-night conditions for sporulation initiation.
Sporulation__MP_min	SMMI	4.3106	5.1741	hours	VPD threshold representing Minimum cumulative wet-night hours required to trigger sporulation.
Sporulation__Tmin	STMI	13.9426	11.8061	°C	Lower temperature limit for sporulation response.
Sporulation__Topt	STOP	25.4161	19.3980	°C	Optimum temperature for sporulation intensity.
Sporulation__Tmax	STMX	35.2136	35.0492	°C	Upper temperature limit for sporulation response.
Dispersal__rain_min	DIRM	12.4525	12.3417	mm	Rainfall threshold triggering rain-driven sporangial dispersal.
Dispersal__wind_min	DIWM	6.4975	6.2638	m/s	Wind-speed threshold triggering wind-driven sporangial dispersal.
Deposition__rain_min	DERM	5.1413	6.0721	mm	Rainfall threshold controlling deposition/washout of airborne propagules.
Zoospore_alive__rain	ZARA	4.0835	5.3267	mm	Rainfall condition threshold used in zoospore survival gating.
Zoospore_alive__vpd	ZVPD	1.9396	1.9674	kPa	Vapor pressure deficit threshold used in zoospore survival gating.
Infection__infection_condition	IC	25.4409	81.4294	/	Infection-establishment requirement threshold.

Parameter	Abbreviation	FSIM	FSIM-S	Unit	Description
Incubation__spl_spores	ISED	1.1063	0.9361	/	Cumulative incubation progress threshold for visible oil-spot onset.
Thresholds__min_spores	TMSO	0.0431	0.1016	/	Minimum viable spore amount retained for process continuity.
Spore__alpha_pos	SAP	/	1.2849	/	Positive spore-pressure coefficient; decreases the effective infection-establishment threshold under favourable spore pressure.
Spore__alpha_neg	SAN	/	0.6306	/	Negative spore-pressure coefficient; increases the effective infection-establishment threshold under suppressive spore pressure.

Model calibration and evaluation

To assess the temporal predictive capability for first visible symptom onset and the applicability of the *First Seasonal Infection Model* (FSIM) and its spore-constrained extension (FSIM-S) under humid subtropical vineyard conditions, parameter calibration and model evaluation were conducted within a site–year-based cross-validation framework. The two models used the same weather and phenology inputs and the same interval-based symptom-onset metric; FSIM-S additionally used the fixed interval-rate, z-score spore-pressure field and the calibrated coefficient β_{sp}.

Calibration strategy

Calibration used a site–year leave-one-out design. In each of the six outer folds, one site–year was held out and the remaining five were used for calibration. Six shared FSIM process parameters were first fitted with a bounded finite coarse-to-fine coordinate search: one-at-a-time screening on the prescribed coarse grids was followed by coarse coordinate descent and local fine coordinate descent. Fine candidates were generated at the parameter-specific step size within two steps on either side of the current value. The calibrated parameters were the infection-establishment degree-hour threshold, minimum wind speed, effective deposition rainfall threshold, sporulation relative-humidity threshold, minimum moist-hours requirement, and heavy-rain wash-off threshold.

After the FSIM parameter vector was fixed, β_{sp} was calibrated on the same five training site–years. The coefficient was bounded to $[0, 1]$ and selected by a coarse grid followed by local refinement with a step of 0.05. Candidate configurations were ranked by penalized MAE; ties were resolved by preferring non-positive mean signed error, then

the smallest absolute signed error, and finally the configuration closest to the prior values.

Performance metrics

Visible disease-onset observations were interval-censored by the survey schedule. For site–year i, let t_i^- be the last disease-negative survey date and t_i^+ the first disease-positive survey date. The observed onset window was therefore

$$O_i = [t_i^-, t_i^+], \qquad t_i^- = t_i^+ - 3\,\mathrm{d}. \tag{7}$$

For each model, $\widehat{O}_i = [\hat{t}_i^-, \hat{t}_i^+]$ was defined as the first complete simulated symptom-onset window. The signed interval distance, expressed in days, was

$$e_i = \begin{cases} 0, & \hat{t}_i^- \le t_i^+ \text{ and } \hat{t}_i^+ \ge t_i^-, \\ \hat{t}_i^- - t_i^+, & \hat{t}_i^- > t_i^+, \\ \hat{t}_i^+ - t_i^-, & \hat{t}_i^+ < t_i^-. \end{cases} \tag{8}$$

Thus, overlapping windows had zero error, whereas non-overlapping windows were assigned the signed distance between their nearest boundaries. For the set C of site–years with a complete predicted window, interval-based MAE and RMSE were calculated as

$$\mathrm{MAE} = \frac{1}{|C|}\sum_{i\in C}|e_i|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{|C|}\sum_{i\in C} e_i^2}. \tag{9}$$

If no complete symptom-onset window was simulated, the site–year received a fixed 60-day penalty in the calibration objective. The corresponding penalized MAE was $\{\sum_{i\in C}|e_i| + (n - |C|) \times 60\,\mathrm{d}\}/n$, where n is the total number of site–years. Complete-window MAE, RMSE, signed mean error, and window coverage were retained for evaluation.

Cross-validation and robustness assessment

Each held-out site–year was evaluated once with the calibrated FSIM parameter vector and once with the same vector augmented by the calibrated β_{sp} and the spore-pressure dose. No observations from the held-out site–year were used during either calibration stage. Cross-validation summaries were based on the held-out interval errors and reported complete window coverage together with interval MAE, RMSE, and signed mean error for FSIM and FSIM-S.

Selection of global parameter configuration

To obtain a single deterministic model configuration for site–year simulations, the fold-specific best parameter sets (those achieving the lowest validation MAE within each fold) were evaluated on the full dataset. The configuration yielding the lowest overall MAE was selected as the final global parameter set. Direct averaging of parameters across folds was avoided, as it led to substantial error inflation due to the non-convex nature of the objective function and the presence of multiple near-optimal parameter regions.

Regional simulation and disease-management scenario analysis

To evaluate the applicability of the FSIM-S model at the regional scale and its potential value for risk-based disease management, regional simulations were conducted across Guangxi Province for the period 2000–2024. Given the limited spatial coverage of field observations and the lack of a spatially explicit vineyard census layer, the regional analysis was framed as a scenario-based assessment rather than as a complete mapping of actual vineyard distribution. Three cultivar scenarios were constructed to ensure comparability across the region, assuming that vineyards were uniformly planted with Wink, Yeniang No. 2, or Kyoho cultivars, respectively. The model was driven by the hourly meteorological dataset described previously (Fig 3).

Considering that growers in the region commonly adopt similar preventive spraying schedules through peer-to-peer technical exchange, chemical disease management was represented using a schedule derived from the 2022 Du'an (Hechi) experimental vineyard (Fig 7) (Chavas and Nauges, 2020; Huffman, 2020; Norton and Alwang, 2020). The potential to avoid early preventive treatments was evaluated relative to this baseline schedule, whereby any preventive fungicide application conducted more than seven days before the simulated first symptom onset was classified as potentially avoidable. The seven-day buffer was used to remain consistent with the protection window commonly expected for vineyard fungicide applications and to avoid classifying near-risk sprays as redundant. Based on this criterion, the multi-year average number of potentially avoidable early applications and the corresponding reduction proportion were calculated.

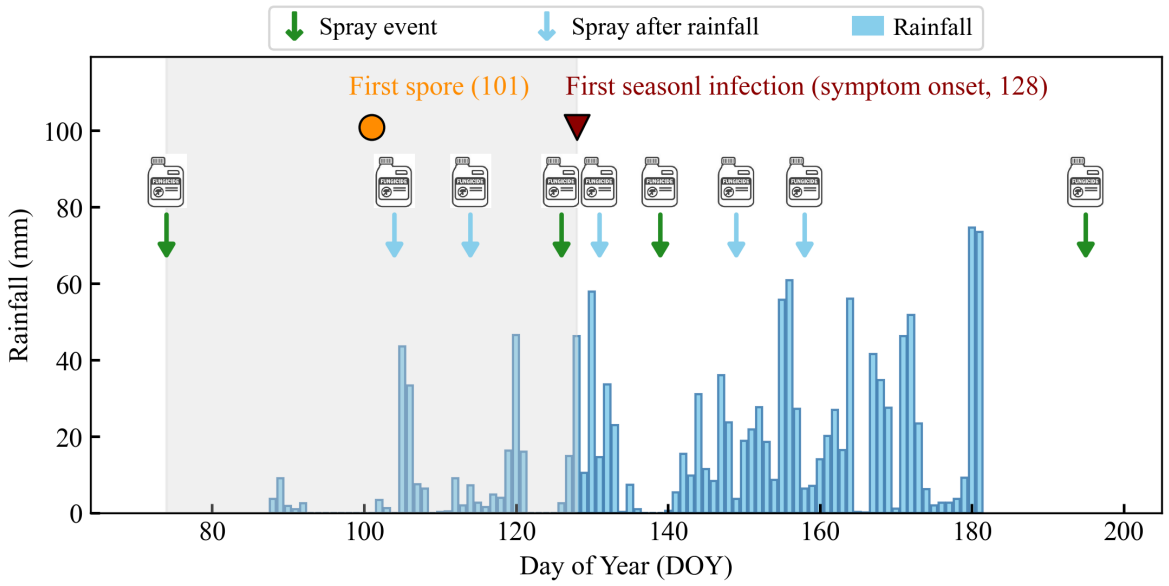


Figure 7: Illustration of the empirically executed fungicide application pattern, documented by field investigators immediately after each grower-implemented spray event in the experimental vineyard in Du'an, Hechi City, during the 2022 growing season. The management practice was characterized by preventive fungicide applications conducted after rainfall events and before the onset of first visible symptoms. The shaded region indicates the corresponding pre-symptomatic period of rainfall-triggered preventive sprays.

Phenology simulation

To provide phenological inputs for regional simulations, we trained a temperature-driven grapevine phenology model from the historical phenology observations. The model follows the growing degree day (GDD) approach commonly used for grapevine phenology (Wang et al., 2020). For each day d, daily GDD was calculated as

$$GDD_d = \max(\overline{T}_d - T_b, 0), \tag{10}$$

where $\overline{T}_d$ is the daily mean air temperature and T_b is the base temperature. Starting on 1 January, thermal time was accumulated between consecutive stages as

$$C_s(t) = \sum_{d=t_{s-1}+1}^{t} GDD_d, \qquad t_s = \min\{t : C_s(t) \geq H_s\}, \tag{11}$$

where t_s is the date of stage s and H_s is its calibrated thermal requirement. The model generated dates for BBCH09, BBCH53, BBCH65, and BBCH81; BBCH53 was derived from the interval between BBCH09 and BBCH65 using the calibrated auxiliary-stage rule. These dates were used as phenological inputs to the regional FSIM-S simulations.

Reconstruction of spore pressure fields

Because continuous airborne spore observations were unavailable for the historical period (2000–2024), while the FSIM-S model requires inoculum pressure information, a meteorology-driven spore-pressure reconstruction module was developed. Field spore monitoring data collected during 2022–2024 were used for parameter calibration, after which spatially explicit spore-pressure time series were reconstructed to drive historical simulations.

The module follows a "potential inoculum reservoir–pressure index" dynamic framework, in which meteorological effects on spore production and loss are represented as source and loss terms:

$$A_t = \max\left(0, \rho A_{t-1} + Source_t - Loss_t\right) \tag{12}$$

where A_t denotes the latent inoculum state. After logarithmic normalization and smoothing, the resulting daily spore-pressure index was used to construct the forcing field for historical FSIM-S simulations.

To ensure consistency with the disease process model, meteorological drivers used in pressure reconstruction shared the same physical interpretation as those controlling infection processes in FSIM-S. The pressure module employed a simplified inoculum-reservoir dynamic that was independently calibrated, enabling forcing input under conditions lacking long-term spore observations. Calibration was performed using Bayesian optimization with cross-validation, and the resulting parameters were applied to reconstruct gridded spore-pressure fields for 2000–2024, which were subsequently coupled to FSIM-S through the spore-pressure dose pathway.

Results

Field observations of disease onset and spore dynamics

Across the six site–years, observed disease-onset windows spanned 19–22 April (Nanning 2024) to 27–30 May (Guilin 2022). Spore-positive collections recorded up to the corresponding onset dates spanned 30 March (Nanning 2023) to 30 May (Guilin 2022), with counts of 1–84 per collection (Fig 8). The earliest first disease-positive assessment occurred in Nanning 2024 (22 April), whereas the latest occurred in Guilin 2022 (30 May); the earliest and latest first spore-positive observations were in Nanning 2023 (30 March) and Guilin 2022 (17 April), respectively. The final spore-positive collection coincided with the first disease-positive assessment in four site–years, whereas it preceded that assessment by 9 days in Hechi 2022 and 3 days in Nanning 2023.

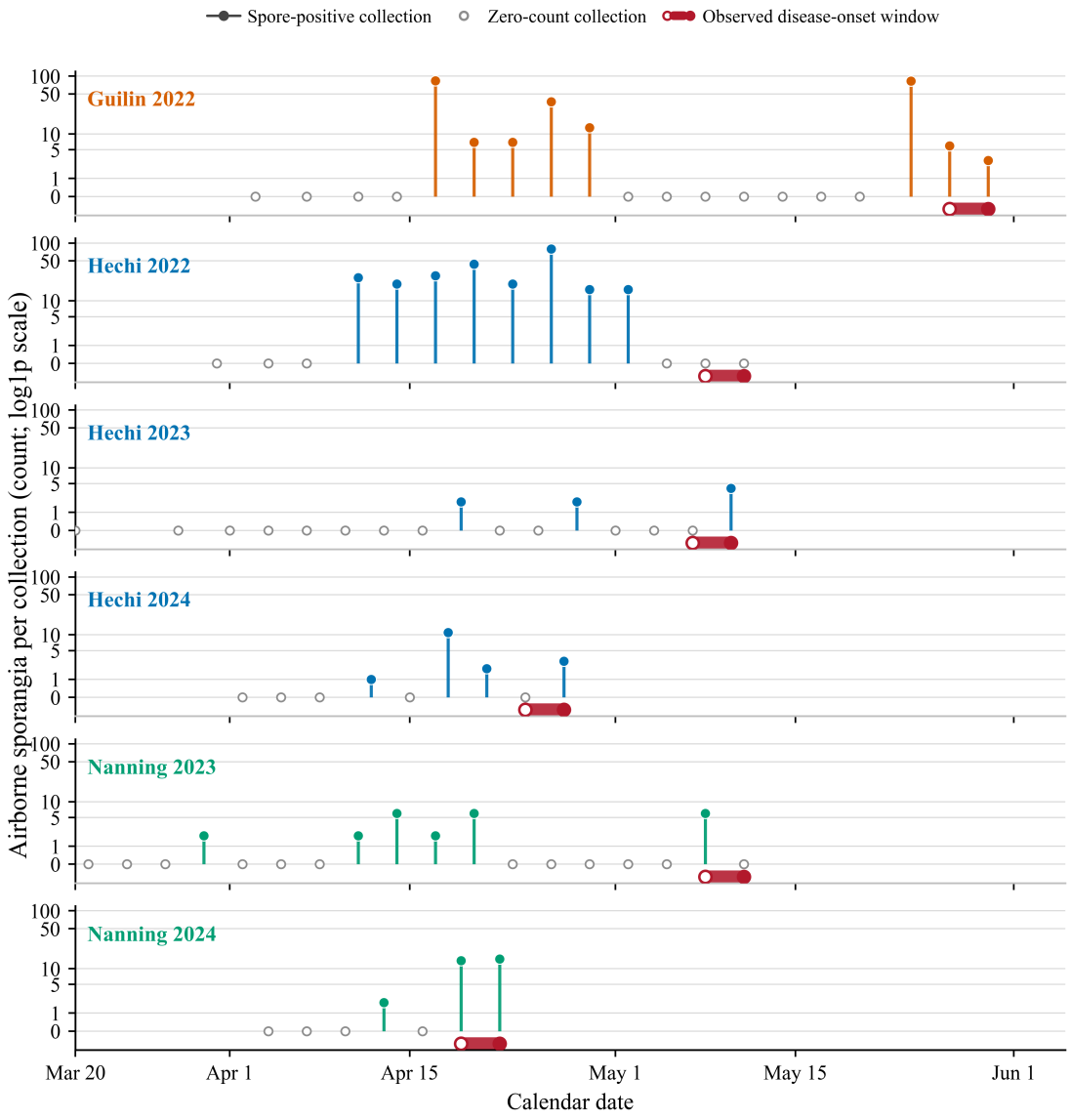


Figure 8: Experimental observations from Guangxi. Spore-positive and zero-count collections are shown on a common calendar axis; red bars denote disease-onset windows between the last negative and first positive assessments.

Sensitivity analysis

The total-order sensitivity ranking was EDR > SRH > MMH > HWR > IDH > WTH (Fig 9). EDR was the most sensitive parameter, whereas WTH had zero total-order sensitivity; SRH, MMH, HWR, and IDH had intermediate sensitivities. This ranking was unchanged across the three Sobol sample sizes (N = 256, 512, and 1024), with only small changes in the corresponding S_T values.

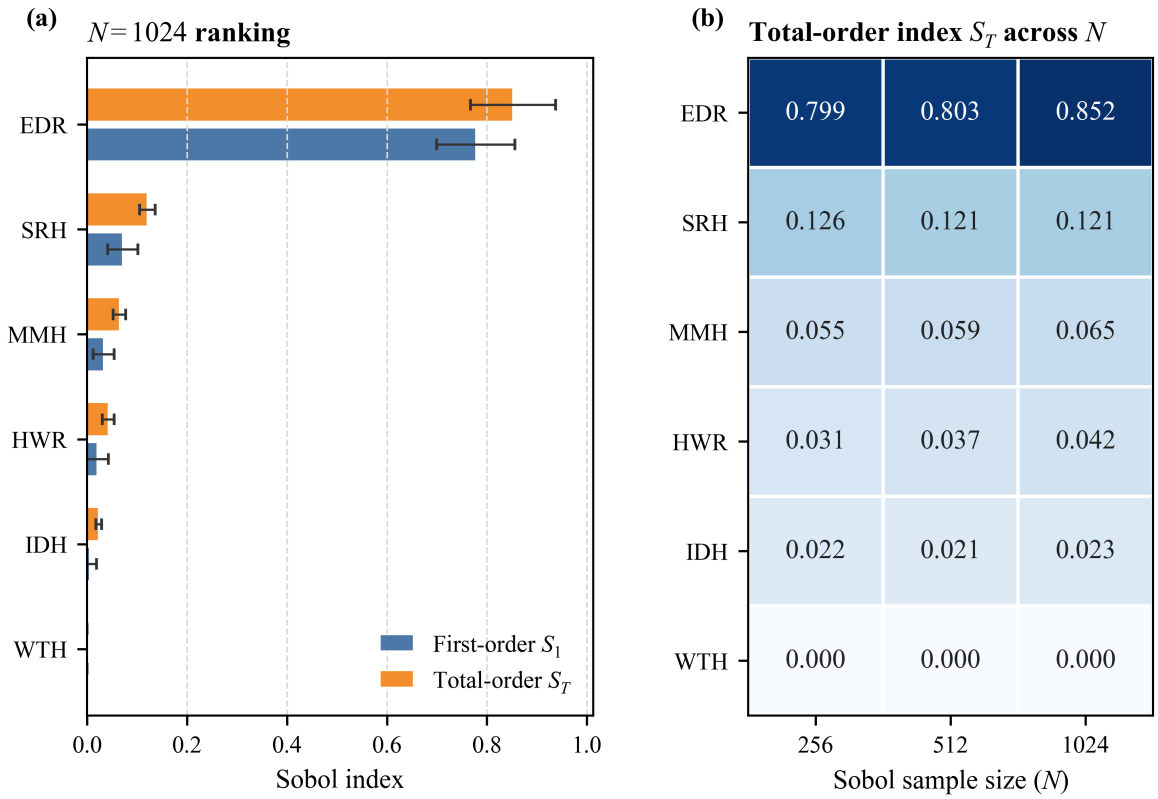


Figure 9: Sobol sensitivity of six FSIM parameters for the censored mean first symptom-onset day. (a) First-order (S_1) and total-order (S_T) indices at N = 1024; (b) total-order indices across N = 256, 512, and 1024. Parameter abbreviations: EDR = effective deposition rainfall threshold; SRH = sporulation relative-humidity threshold; MMH = minimum moist-hour threshold; HWR = heavy-rain wash-off threshold; IDH = infection degree-hour threshold; WTH = minimum wind-speed threshold.

Model performance

FSIM had a training-set MAE of 5.67 days and RMSE of 7.18 days, and a validation-set MAE of 5.67 days and RMSE of 7.26 days (Fig 10).

FSIM-S had a training-set MAE of 3.83 days and RMSE of 5.99 days, and a validation-set MAE of 3.83 days and RMSE of 6.18 days. Relative to FSIM, MAE decreased by 32.4% in both sets; RMSE decreased by 16.6% in the training set and 14.9% in the validation set.

From FSIM to FSIM-S, the city–year errors changed as follows: Duan 2024, 0 to 0 days; Duan 2023, -6 to +4 days; Duan 2022, -14 to -14 days; Mingyang 2024, +4 to +4 days; Mingyang 2023, +2 to +1 days; and Guilin 2022, -8 to 0 days (Fig 11). Zero error occurred for Duan 2024 under both models and for Guilin 2022 under FSIM-S. The largest error magnitude was 14 days in Duan 2022 and remained unchanged between models.

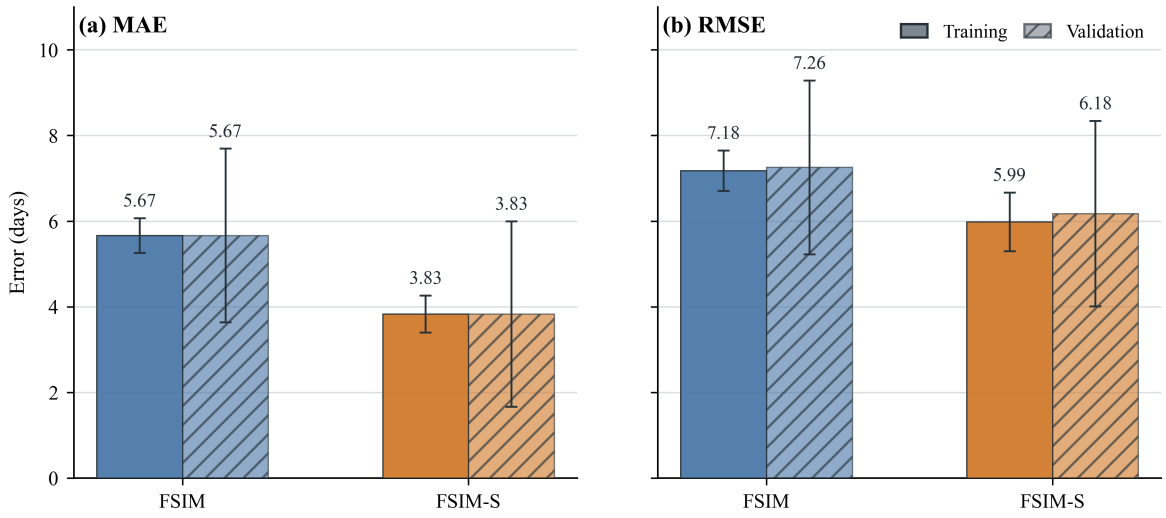


Figure 10: Aggregate first symptom-onset errors for FSIM and FSIM-S. (a) Mean absolute error (MAE); (b) root mean square error (RMSE). Bars distinguish training and validation sets; values are in days.

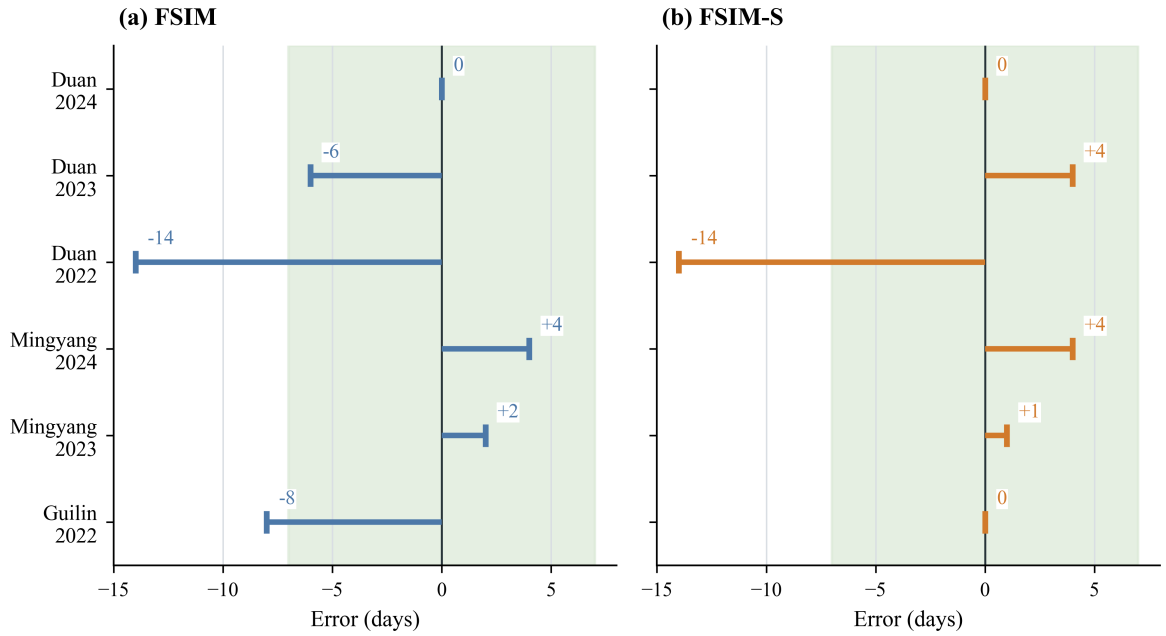


Figure 11: City–year errors in first symptom-onset timing for FSIM and FSIM-S. The shaded region indicates errors within ±7 days of the observed date.

Spatial distribution of first symptom onset across Guangxi (2000–2024)

Based on historical ERA5 meteorological data, the spatial distribution of first symptom onset across Guangxi from 2000 to 2024 was reconstructed using FSIM-S, simulated under three variety-specific phenology scenarios: Wink (WK), Kyoho (JF), and Yeniang No. 2 (MPT) (Fig 12).

Multi-year mean onset dates were similar across varieties, with WK and JF averaging DOY 195 and MPT DOY 198 (median: 196, 196, and 199, respectively). Spatially, earlier onset dates were consistently simulated in central–southern and parts of southwestern Guangxi, whereas later onset occurred in northern and northeastern areas (Fig 12a–c). This spatial gradient was consistent across all three varieties.

Frequency distributions (Fig 12d) showed that the majority of onset dates fell within DOY 180–210 across all varieties, with peak frequencies concentrated around DOY 190–200. The overall distribution structures were highly similar among varieties.

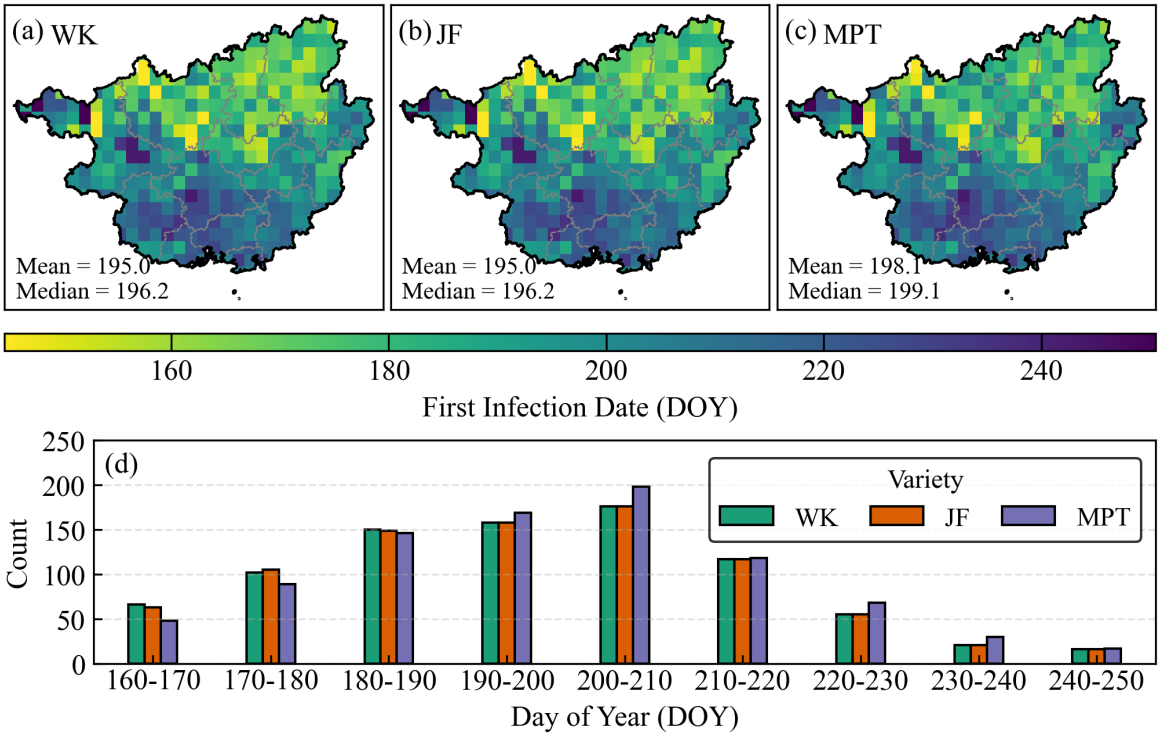


Figure 12: Simulated mean onset dates (day of year, DOY) of first visible symptoms under the FSIM-S model for three varieties: (a–c) spatial distributions of the Wink (WK), Kyoho (JF), and Yeniang No. 2 (MPT), and (d) their DOY frequency distributions.

Regional disease-management implications (2000–2024)

The potential to improve early disease-management timing across Guangxi from 2000 to 2024 was evaluated by comparing the FSIM-S-derived timing of first visible symptom onset with a representative conventional fixed-calendar programme (Fig 13). Across the three grapevine varieties, the model-guided strategy identified approximately three potentially avoidable early preventive applications per season on average (mean: 3.0 sprays; median: 2.7 sprays), with minimal variation among varieties. This result indicates that the main management value of FSIM-S lies in delaying unnecessary pre-risk sprays while retaining protection near the simulated symptom-onset window.

Spatially, higher saving potential was concentrated in central–southern and southwestern Guangxi, where simulated symptom onset tended to occur later relative to the representative calendar programme, whereas northern and north-eastern areas showed relatively lower savings (Fig 13a–c). This spatial pattern was consistent across all three varieties.

The frequency distribution (Fig 13d) showed that spray savings were predominantly two to four applications per season, with two or three sprays being the most common. Higher savings (≥5 sprays) occurred less frequently but were present in certain locations. The overall distribution structures were highly similar among varieties, suggesting that spatial climatic variability exerted a stronger influence on early-season management timing than the simplified variety scenarios considered here.

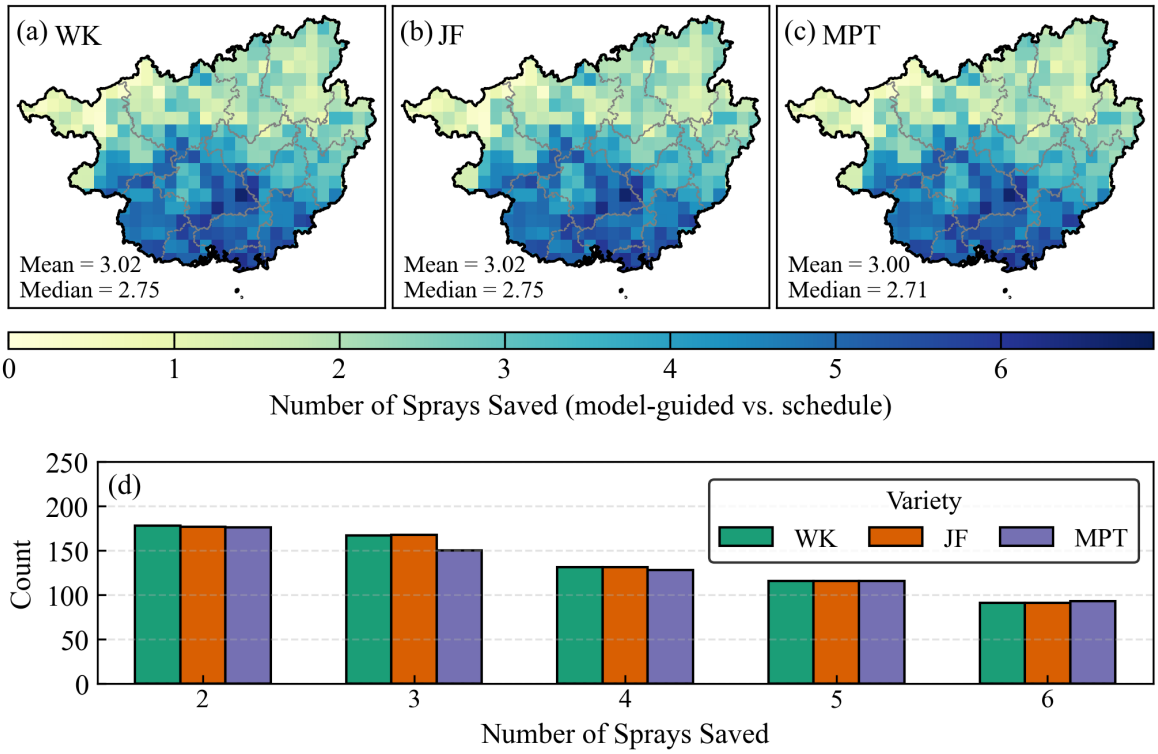


Figure 13: Spatial distribution and frequency of model-guided spray savings under the FSIM-S model: (a–c) multi-year mean number of sprays saved per grid cell for the Wink (WK), Kyoho (JF), and Yeniang No. 2 (MPT), and (d) frequency distribution of spray savings.

Discussion

This study addressed a key agronomic challenge in subtropical viticulture: improving early-season grapevine downy mildew management when viable inoculum may be present before weather and crop conditions become jointly favourable for infection establishment. By reconstructing Rossi's primary and secondary infection formulations into a First Seasonal Infection Model (FSIM), and then incorporating observation-based spore-pressure dynamics (FSIM-S),

the study linked on-farm inoculum monitoring with agroclimatic modelling of symptom onset.

Across multiple site–year datasets from Guangxi, China, FSIM-S improved prediction accuracy and stability compared with FSIM while retaining the dominant influence of meteorological drivers. These results support the use of FSIM-S as a region-adapted mechanistic framework for field-based downy mildew decision support in subtropical vineyards, rather than as an empirical model fitted only to symptom dates.

Inoculum dynamics and the drivers of first seasonal symptom onset

In temperate systems, seasonal epidemics are typically initiated by overwintering oospores after a distinct climatic break. In contrast, the continuous presence of airborne sporangia during the early growing season (Fig 5, Fig 8) indicates that inoculum in the studied subtropical vineyards is not restricted to a discrete primary infection event. This distinction has direct implications for model structure, because frameworks centred on oospore-driven primary infection may not fully represent systems in which asexual inoculum is recurrently available.

Despite this continuous inoculum presence, the timing of first visible symptom onset varied substantially across sites and years (Fig 8). This variability is consistent with interannual differences in early-season meteorological conditions. Temperature, relative humidity, and leaf wetness duration regulate infection establishment (Rossi and Caffi, 2012; Yi et al., 2019; Yu et al., 2022), whereas sustained rainfall can promote sporangial production, release and dispersal from vineyard inoculum sources (Orlandini et al., 2008). Accordingly, climate variability affects disease onset through both infection-establishment suitability and inoculum dynamics, explaining why first symptom onset cannot be inferred from weather suitability alone.

The temporal decoupling between sporangia detection and symptom onset provides further evidence for this interpretation. Across all datasets, the interval between first airborne sporangia detection and subsequent symptom appearance ranged from 9 to 72 days (Fig 8), indicating that the presence of inoculum is necessary but insufficient for infection establishment and subsequent symptom expression. Infection establishment depends also on host phenological susceptibility (Hui et al., 2022) and on sufficient inoculum pressure to support effective deposition and infection establishment (Scholthof, 2007; Van der Plank, 2013; Zhang et al., 2025). This decoupling indicates that inoculum pressure should be represented as a dynamic constraint on infection triggering, rather than treated as a static prerequisite.

Model enhancement

The incorporation of spore-pressure dynamics into FSIM-S improved predictive performance relative to the baseline FSIM, with validation MAE reduced from 5.67 to 3.83 days (32.4% reduction) (Fig 10). The improvement suggests that first symptom onset tends to occur within comparatively narrow windows in which environmental suitability

coincides with sufficient pathogen pressure, consistent with the disease triangle concept (Van der Plank, 2013). The model therefore adds process information beyond weather-only infection suitability while retaining direct biological interpretability.

City–year errors showed the largest error magnitude in Duan 2022 (14 days; error = -14 days) for both models, whereas Duan 2024 had zero error under both models and Guilin 2022 had zero error under FSIM-S (Fig 11). FSIM-S reduced the absolute error in Duan 2023, Mingyang 2023, and Guilin 2022, while the remaining city–years were unchanged.

Overall, FSIM provides a mechanistic representation of first seasonal infection-to-symptom processes in subtropical vineyards where recurrent sporangia production is a major inoculum source. FSIM-S further integrates an observation-derived spore-pressure field, improving responsiveness to symptom-onset timing variability while preserving the core biological processes inherited from mechanistic infection modelling. This balance between biological structure and field observation is important for developing transferable early-warning tools under variable subtropical conditions.

Implications for inoculum-informed preventive timing

Airborne inoculum monitoring can complement weather-based assessment by indicating the background pathogen pressure under which favourable infection conditions occur. FSIM-S connects these observations with a mechanistic sequence, providing a basis for investigating the timing of early-season risk. The evaluated endpoint is visible symptom onset, whereas preventive protection concerns the earlier infection-risk period. Reconstructed symptom-onset timing therefore requires prospective evaluation with inputs available at the time of a decision before they can be translated into treatment recommendations. Demonstrating reduced fungicide use with maintained disease control would additionally require field comparisons of conventional, model-guided and untreated programmes.

Agronomic implications for climate-smart vineyard disease management

The regional simulations provide an agronomic interpretation of when model-guided protection would be most relevant across Guangxi. FSIM-S predicted a mean first symptom-onset date between DOY 195 and 198, with a high-frequency window spanning DOY 180–210 (Fig 12). This period overlaps with the regional flowering–fruit-set stage, when young tissues are particularly susceptible (Wang et al., 2020; Cheng et al., 2022), and with the Meiyu season, characterized by warm, humid conditions and intermittent rainfall (Song et al., 2021; Tian et al., 2022). Thus, the regional results are not only spatial reconstructions of symptom-onset timing; they also identify the seasonal context in which downy mildew decisions are most consequential for vineyard production.

Within this context, the management implication of FSIM-S is to translate infection biology and agroclimatic variability into a time-explicit risk signal. This signal can support decisions on early scouting, spore monitoring intensity,

protection timing and chemical intervention. When model outputs were compared with conventional calendar-based programmes, the model-guided scenario avoided an average of approximately three early preventive sprays per season (range: 2–5), equivalent to a 20–50% reduction in the simulated early-season programme (Gisi and Sierotzki, 2008; Rossi et al., 2008; Rossin et al., 2015; Koledenkova et al., 2022). This result should be interpreted as a timing benefit rather than as a generic fungicide-reduction claim: the model delays pre-risk sprays while retaining protection near the simulated symptom-onset window.

The climate-smart value of this framework therefore lies in shifting subtropical downy mildew management from fixed calendar schedules toward risk-informed decision support. FSIM-S does not imply that early protection can be omitted; rather, it identifies periods when host susceptibility, spore-pressure dynamics and agroclimatic suitability converge. Such information can improve the biological justification of interventions, increase pesticide-use efficiency, and support more sustainable viticulture under humid subtropical conditions (Russell, 1995; Zubrod et al., 2019; Zhang et al., 2020).

Limitations and future directions

First, model parameterization used observations from two commercial vineyards and one experimental orchard over three growing seasons. This provides multi-site and multi-year evidence under realistic subtropical production conditions, while additional observations across more regions, cultivars and management systems would further broaden the calibration domain. The 2022 Guilin season, for example, provided a useful contrast because unusually low early-season humidity and irregular rainfall (Fig 3) may have delayed or suppressed infection establishment. Such contrasting cases are valuable for testing model behaviour, but wider datasets would support more explicit cultivar- or region-specific parameterization (Bellin et al., 2009; Gessler et al., 2011; Koledenkova et al., 2022).

Second, FSIM-S currently uses a deterministic mechanistic structure to describe the coupling between meteorological conditions and spore-pressure dynamics. This formulation favours stability, interpretability and transparent biological assumptions. Future probabilistic extensions could complement this structure by quantifying uncertainty associated with rainfall variability, spore sampling error and fine-scale atmospheric heterogeneity, particularly when the model is applied to operational warning systems or climate scenario analysis.

Third, the regional analysis was designed as a scenario-based assessment rather than a vineyard-by-vineyard operational map. It therefore used simplified assumptions about vineyard distribution, canopy structure and cultivar allocation. These assumptions are appropriate for evaluating broad regional patterns and management windows, but finer-scale implementation would benefit from vineyard land-use layers, cultivar distribution data, remote-sensing information and canopy microclimate observations (Tesic et al., 2007; Perria et al., 2022). Such additions would extend the framework

from regional risk assessment toward more spatially resolved disease-management support.

Finally, the six site–years provide a limited basis for geographical or cultivar transfer because site and cultivar are confounded. Further independent observations and prospective evaluation are needed to establish the applicability of the framework beyond these vineyards.

Conclusion

FSIM links established cohort-based and airborne-sporangium processes under a recurrent-airborne-inoculum premise to reconstruct first visible downy mildew symptom onset in subtropical vineyards. FSIM-S incorporates field-observed spore pressure within the same process chain, enabling assessment of its incremental information for symptom-onset estimation. Multi-year and multi-site validation demonstrated that FSIM-S reproduced first visible symptom-onset timing with a mean absolute error of 6.3 days and a root mean square error of 6.5 days. Long-term regional simulations (2000–2024) captured spatial patterns of symptom onset and indicated that model-guided timing could avoid 2–5 early preventive sprays per season compared with a conventional calendar-based programme. Overall, the framework connects inoculum monitoring with infection biology and provides a basis for climate-smart, model-guided disease management under subtropical conditions.

CRediT authorship contribution statement

Linjia Yao: Investigation, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. Bin Chen: Investigation, Writing – original draft, Formal analysis. Writing -review & editing. Vittorio Rossi: Formal analysis, Writing -review & editing. Ling Yin: Funding acquisition, Writing -review & editing. Shumei Wei: Investigation. Qi Tian: Investigation. Changqing Yan: Writing -review & editing. Hui Wang: Writing -review & editing. Antonio Fernandes Nogueira Júnior: Writing -review & editing. Yu Huang: Investigation. Qiang Yu: Conceptualization, Supervision. Junjie Qu: Funding acquisition. Gang Zhao: Software, Methodology, Conceptualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

The model structure, code, and data supporting the findings of this study are publicly available at: https://github.com /SmartAG-Team/grape_disease.git. The repository includes (i) field observations of first visible symptom onset, (ii) airborne sporangia monitoring data, (iii) site–year meteorological data used for model calibration and validation, and (iv) code for model implementation, calibration, and figure generation. Additional materials may be available from the corresponding author upon reasonable request (gang.zhao@nwafu.edu.cn (Gang Zhao)).

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version.